Machine Learning Syllabus

Python For AI & ML

1. Python Basics
2. Python Functions, Packages & Routines
3. Jupyter Notebook – Installation & Function
4. Pandas, NumPy, Matplotlib, Seaborn
5. Working with Data Structures, Arrays, Vectors & Data Frames

Statistical Learning

1. Descriptive Statistics
2. Probability & Conditional Probability
3. Hypothesis Testing
4. Inferential Statistics
5. Probability Distribution – Types of distribution – binomial, Poisson & normal distribution

Supervised Learning

1. Multiple Variable Linear Regression
2. Logistic Regression
3. Naïve Bayes Classifiers
4. Multiple Regression
5. K-NN Classification
6. Support Vector Machines

Unsupervised Learning

1. K-Means Clustering
2. High-Dimensional Clustering
3. Hierarchical Clustering
4. Dimension Reduction-PCA

Ensemble Techniques

1. Decision Trees
2. Bagging
3. Random Forests
4. Boosting

Recommendation Systems

1. Introduction to recommendation system
2. Popularity based model
3. Hybrid Models
4. Content Based recommendation system
5. Collaborative filtering

Neural Network Basics

1. Gradient Descent
2. Batch Normalization
3. Hyper Parameter tuning
4. Activation and Loss function

5. Deep Neural Networks

6. Intro to Perceptron & Neural Networks

Computer Vision

1. Intro to Convolutional Neural Networks
2. Convolution,Pooling,Padding&Mechanism
3. Transfer Learning
4. Forward propagation & Backpropagation for CNNs
5. CNN architectures like AlexNet, VGGNet, InceptionNet & ResNet

Statistical NLP

1. Bag of Words Model
2. POS Tagging
3. Tokenization
4. Word Vectorizer
5. TF-IDF
6. Named Entity Recognition
7. Stop Words

Sequential NLP

1. Intro to Sequential data
2. Vanishing & Exploding gradients in RNNs
3. LSTMs
4. GRUs – Gated recurrent unit
5. Case study: Sentimental analysis
6. RNNs and its mechanisms
7. Time series analysis
8. LSTMs with attention mechanism
9. Case study: Machine Translation

Languages & Tools

1. Python
2. Python ML library scikit-learn
3. NLP library NLTK
4. Keras
5. Data Libraries like Pandas, Numpy, Scipy
6. Python visualization library Matplotlib
7. Tensor Flow
8. Seaborn

Advanced Computer Vision

1. Semantic Segmentation
2. Siamese Networks
3. YOLO
4. Object & face recognition

GANs

1. GAN - Generative Adversarial Net Intro
2. How GANs work?
3. Auto Encoders
4. Applications of GANs

Reinforcement Learning

1. Value based methods Q-Learning
2. Policy based methods